

LAB 05

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import numpy as np
import matplotlib.pyplot as plt
from collections import Counter

np.random.seed(42)
data = np.random.rand(100)

train_data = data[:50]
train_labels = ["Class1" if x <= 0.5 else "Class2" for x in train_data]

test_data = data[50:]

def euclidean_distance(x1, x2):
    return abs(x1 - x2)

def knn_classifier(train_data, train_labels, test_point, k):

    distances = []
    for i in range(len(train_data)):
        dist = euclidean_distance(test_point, train_data[i])
        distances.append((dist, train_labels[i]))

    distances.sort(key=lambda x: x[0])
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k_nearest_neighbors = distances[:k]
k_nearest_labels = [label for _, label in k_nearest_neighbors]

return Counter(k_nearest_labels).most_common(1)[0][0]

k_values = [1, 2, 3, 4, 5, 20, 30]
results = {}

print("--- k-Nearest Neighbors Classification ---")

for k in k_values:
    print(f"\nProcessing for k = {k}...")
    classified_labels = [knn_classifier(train_data, train_labels, tp, k) for tp in test_data]
    results[k] = classified_labels

    for i in range(5):
        print(f"Point x{i+51} (value: {test_data[i]:.4f}) -> {classified_labels[i]}")

for k in k_values:
    classified_labels = results[k]

    class1_test = [test_data[i] for i in range(len(test_data)) if classified_labels[i] == "Class1"]
    class2_test = [test_data[i] for i in range(len(test_data)) if classified_labels[i] == "Class2"]

plt.figure(figsize=(10, 2)) # Shorter height since it's 1D data

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train_colors = ["blue" if l == "Class1" else "red" for l in train_labels]

plt.scatter(train_data, [0] * len(train_data), c=train_colors, label="Train Data", marker="o",
alpha=0.5)


if class1_test:
    plt.scatter(class1_test, [1] * len(class1_test), c="blue", label="Class1 (Test)", marker="x")
if class2_test:
    plt.scatter(class2_test, [1] * len(class2_test), c="red", label="Class2 (Test)", marker="x")


plt.title(f"k-NN Results for k = {k}")
plt.yticks([0, 1], ["Train", "Test"])
plt.xlabel("Value of x")
plt.legend(loc='center left', bbox_to_anchor=(1, 0.5))
plt.grid(axis='x', linestyle='--')
plt.show()
```